LAMP LLM to ASP Modeling Protocol

Task

ASP rule induction. Given facts F and a target answer set A^*, find a small program Π with

$$AS(F \cup \Pi) = \{A^*\}.$$

Bounded unary normal rules. Humans find this hard.

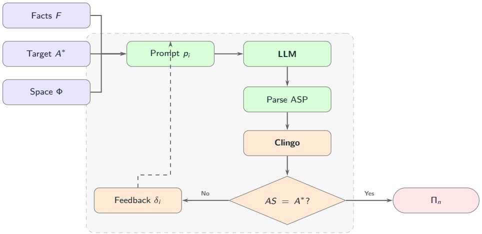
Loop

- LLM proposes rules from F, A^*, schema bounds.
- CLINGO verifies the stable model.
- Mismatches feed back. Up to 3 attempts.
- No fine tuning. Solver decides correctness.

Setup

120 synthetic benchmarks varying predicates, terms, facts, body literals, negation, derived atoms. ILASP baseline with 420 s timeout.

Iterative Loop



Results across 120 benchmarks

System	1st	Final	Fails
gpt-5-mini	90.8%	99.2%	1
gpt-oss	83.3%	98.3%	2
qwen3-coder	19.2%	64.2%	43
ILASP	—	97.5%	3